N A Adarsh Pritam

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Google Scholar

OVERVIEW

I am currently a **Research Intern at the Centre for Brain Research (CBR), Indian Institute of Science (IISc)**, where I primarily work on **machine learning methodologies for dementia research** (for early detection of cognitive decline). My research experience includes **Representation Learning, Generative Models, Computer Vision, Multimodal Learning, NeuroAI and Medical AI**.

RESEARCH EXPERIENCE

J: Journal   C: Conference   W: Workshop   P: Preprint

CBRAIN Research Intern **May 2026 – Present**
Centre for Brain Research (CBR) @ Indian Institute of Science (IISc), Bangalore
Principal Investigator: Dr. Thomas Gregor Issac

- Developed machine learning methods for early identification of **Mild Cognitive Impairment (MCI)** using multimodal longitudinal data from the **TLSA** and **SANSCOG** cohorts ($>$ 10,000 **participants**); improved baseline prediction accuracy ($\approx +15\%$) over statisical baselines through BORUTA feature selection and Bayesian hyperparameter optimization.
- Evaluated **Kaplan–Meier**, **Cox Proportional Hazards**, **Random Survival Forests**, and **DeepSurv** for longitudinal risk prediction; co-authoring a clinical dementia study.

Graduate Research Student **Aug 2025 – May 2026**
Alliance University, Bangalore
Supervisor: Dr. Jeba Shiney

- Proposed a **hierarchical brain-to-latent alignment framework**, demonstrating nonlinear decoders improve semantic alignment ($\Delta \approx +0.47$) with limited gains for diffusion latents ; Mapped **fMRI** signals to **Stable Diffusion** and **CLIP** latent spaces using hierarchical linear and nonlinear decoders, achieving $\approx$ 85% CLIP semantic similarity.
- Benchmarked **StyleGAN2-ADA** and **DDPM** for dermoscopic image augmentation across **14k+ melanoma images** ; Improved melanoma detection by 8–15% absolute F1 (ROC-AUC $\approx$ 0.98) and evaluated synthetic image quality using FID, KID, and downstream classification metrics.

Publications: P1 , P2 , C1

Summer ML Research Intern **May 2025 – Sep 2025**
i2CS Research Group @ Indian Institute of Information Technology (IIIT) Kottayam
Supervisor: Dr. Kala S

- Designed a multimodal plant disease diagnosis framework integrating CLIP (ViT-L) visual representations with InternLM2 language models ; Implemented projection-based vision-language alignment, scalable PyTorch training pipelines, and contributed to experimental evaluation and manuscript preparation.

PUBLICATIONS

J: Journal   C: Conference   W: Workshop   P: Preprint

P1 **Brain2VLM: Hierarchical Alignment Between Cortical Representations and Vision-Language Latent Spaces**
N A Adarsh Pritam, Jeba Shiney O, Sanyam Jain.
bioRxiv Preprint, 2026 (Currently under review).

P2 **SkinGenBench: Generative Model and Preprocessing Effects for Synthetic Dermoscopic Augmentation in Melanoma Diagnosis**
N A Adarsh Pritam, Jeba Shiney O, Sanyam Jain.
arXiv Preprint, 2025 (Currently under review).

C1 **American Sign Language to Fluent English Detection and Translation Using T5**
N A Adarsh Pritam, Asha Kurian.
IEEE International Conference on Emerging Technologies in Computing and Communication (IEEE - ETCC), 2025.

OPEN SOURCE PROJECTS

- **LLaMA LLM Reimplementation** — Reimplemented a decoder-only **LLaMA** architecture from scratch with custom tokenization, transformer layers, and **QLoRA** fine-tuning for personalized conversational modeling. GitHub / Blog
- **PyTorch Classification Extended** — Extended the **bearpaw image classification** framework repository with **Grad-CAM** visualization and expanded architecture support. GitHub

EDUCATION

Alliance University, Bangalore 2024 – 2026
Master of Science in Data Science **CGPA**: 8.6

Bengaluru City University, Bangalore 2024 – 2026
Bachelor of Science in Mathematics and Statistics **CGPA**: 6.5

Relevant Coursework: Machine Learning I & II, Deep Learning, ML Techniques for Image Processing, Mathematical Foundations for Data Science, Natural Language Processing, Inferential and Descriptive Statistics, Foundations of Data Science, Applied Mathematics, Probability, Operations Research, ML-Operations (MLOps), Research Methodology

SKILLS

Programming: Python, R, SQL
Research Areas: Representation Learning, Generative Models, Computer Vision, NeuroAI, Medical AI, Deep Learning, Machine Learning, Statistical Learning
Core Libraries: PyTorch, Torchvision, OpenCV, Transformers, Scikit-learn, Optuna, Shap, Pandas, Numpy
Tools: Git, Docker, MLflow, ZenML, AWS, Weights&Biases, Latex

CERTIFICATIONS

- AIF-C01: AWS Certified AI Practitioner - Amazon Web Services (2024)
- Fundamentals of Deep Learning - NVIDIA (2025)
- Mathematics for Machine Learning and Data Science – DeepLearning.AI (2024)

ACHIEVEMENTS

- Invited to serve on the Board of Studies (BoS) curriculum committee for the MSc Data Science program, contributing to curriculum design for subsequent (2026 onward) student cohorts.
- Invited to deliver seminars on machine learning research methodology for MSc Data Science students.

References

REFERENCES

1. Dr. Thomas Gregor Issac (thomasgregor@cbr-iisc.ac.in), Associate Professor,
Centre for Brain Research, Indian Institute of Science.
2. Dr. Jeba Shiney (jeba.shiney@alliance.edu.in), Professor,
Alliance School of Advanced Computing, Alliance University.
3. Dr. Kala S (kala@iiitkottayam.ac.in), Assistant Professor & Head of i2CS Research Group,
Indian Institute of Information Technology (IIIT) Kottayam.